Computer Science & IT



C Programming

Data Types and Operators

Lecture No. 04



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Recap of Previous Lecture



- Topic — Problem Solving
- Topic — Operator & Expression
- Topic — L-value and R-value
- Topic
- Topic

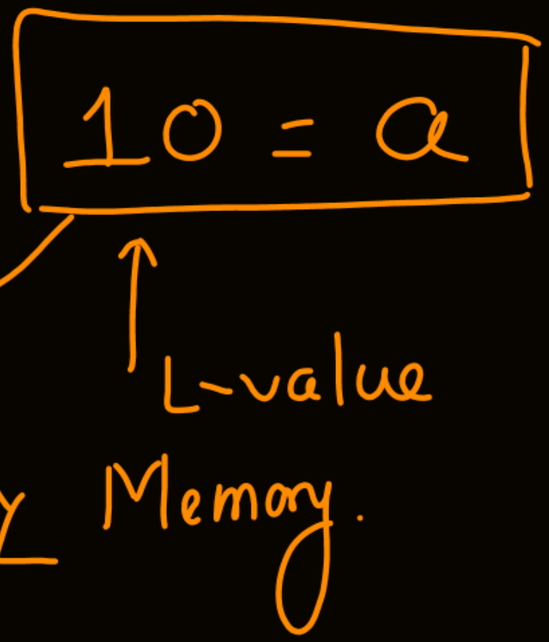
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Topics to be Covered



- Topic — Arithmetic Expression
- Topic — Precedence & Associativity
- Topic — BODMAS
- Topic
- Topic
- Topic

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Memory Management

Operating System



Topic : Question

Consider the following program

```
#include <stdio.h>
int main( ){
      int a = 10;
      20 = a;
}
```

Which of the following TRUE for above program?

(A)Warning

(B)L-value required Error

(C)Location 20 modified by a

(D)No Error

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Topic : Question

a = 20+30
50

Variable

Consider the following program

```
#include <stdio.h>
int main( ){
      int a = 10;
      20+30 = a;
}
```

Which of the following TRUE for above program?

(A)Warning

(B)L-value required ✓ error

(C)Location 20 modified by a
+30

(D)No Error

20+30 → arithmetic Expression

L-value is Not present.



Topic : Question MSQ

MSQ

(B) (C) (D)

Which of the following TRUE for C- program?

(A)L-value of a declared variable is modifiable

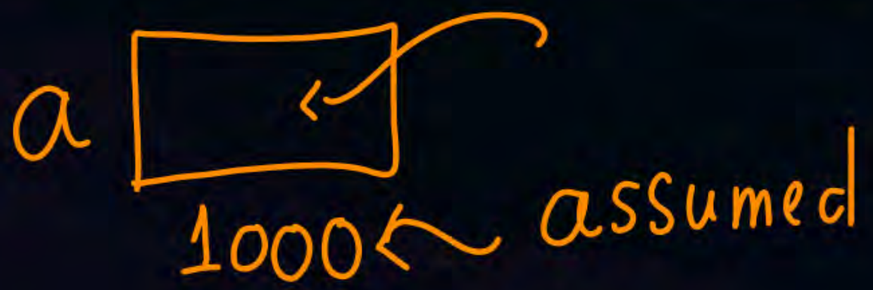
(B) R-value of a variable is modifiable

(C)We can access the L-value of a variable

(D) Expression has R-value associated with them

New value assign

```
a = 20;
a = a + 20;
a = 60;
```



Slide



Topic : Arithmetic Operator

+, −, *, /, %

modulo

Slide



Topic : Arithmetic Operator

Dividend
Divisor
Quotient
Remainder

Dividend = Quotient * Divisor + Remainder

$7 = 2 \times 3 + 1$

2 - Quotient

$3\overline{)7}$ ← dividend

3 ← divisor

2 ← Quotient

7 ← Dividend

6

1 ← ~~Rain~~ Remainder

divisor

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Topic : Arithmetic Operator

7 / 3;	-7/3,	7/-3,	-7/-3
Q : 2	Q -2	Q -2	Q : 2
R : 1	R -1	R 1	R : -1

Divison / - quotient

Modulo % Remainder

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Topic : Arithmetic Operator

7 / 3, -7/3, 7/-3, -7/-3

	7/3	7/-3	-7/3	-7/-3
Dividend	7	-7	7	-7
Divisor	3	3	-3	-3
Quotient	2	-2	-2	2
Remainder	1	-1	1	-1

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Topic : Rule

Dividend posihve then Remainder postive

Dividend Negative Remainder Negative



Program

```
#include <stdio.h>
int main() {
   printf("%d\n",7/3);
   printf("%d\n",7/-3);
   printf("%d\n",-7/3);
   printf("%d\n",-7/-3);
   printf("%d\n",7%3);
   printf("%d\n",7%-3);
   printf("%d\n",-7%3);
   printf("%d\n",-7%-3);
   return 0;
}
```



Arithmetic Operation Between int and float

1. Operation between integer & integer ans integer

$1/2 = 0$, $5/2 = 2$

2. Operation between integer & float will be float

$5/2.0$ or $\frac{5.0}{2} = \underline{2.5}$

3. Operation between float & float answer will be float only

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Precedence Rule

* Order of evaluation of operator decided by precedence Rule..

precedence table

$2 + (3 \times 3)$

$= 11$

BODMAS

order of evaluation

Bracket

order of $\sqrt{x}$, x^2

A - Addition

S - Subtraction

D - Division,

M. multiplication

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Associativity Rule

if operator is having same precedence then order of evaluation decided by Associativity Rule.

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Precedence & Associativity Rule

bracket	( )	Associativity	Highest ✓
Unary minus	-	→ Same precedence	
Multiplicative	*, / , %	Left to Right	
Additive	+ -	Left to Right	
Equal	=	Right to Left	Lowest

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Associativity

$5+6*4 = 5+24 = 29$

precedence

example 1

$5+6-4+2-1$ Left Associative

$11-4=7$

exm 2 $5+6-7+8 = 11-7+8 = 4+8 = 12$

example3: $6/2*3 = 3*3 = 9$

Equal to (=)

```
int a = 10, b = 20, c;
c = (a = b);
```

Right to left

a 20

b 20

c 20

(c = a)

c 20

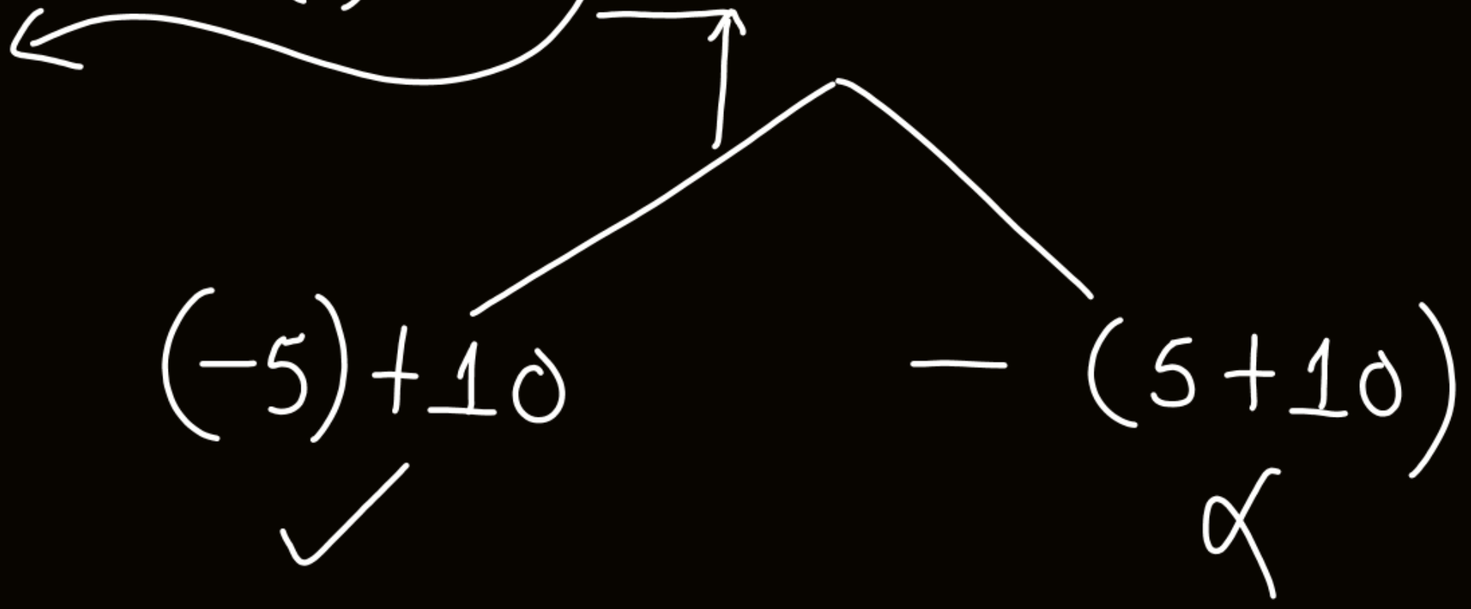
Unary Minus

(1) $-5 * 4 = -20$

(3) $-7/2$

Unary -

(2) $-5 + 10 = 5$





Toipc:Question

```
#include <stdio.h>
int main(void){
   float x;
   x = 7*2.0/2+10/3;
   printf("%f", x);
   return 0;
}
```

The value of x is ___

(a) 10
(b) 10.0
(c) 10.33
(d) 11.0

7 * 2.0 / 2 + 10 / 3

14.0 / 2 + 3

7.0 + 3 = 10.0

Slide



Toipc:Question

```
#include<stdio.h>
   int main(){
   int x;
   x= (-2) + 11 - 7 * 9 % 6 / 12;
   printf("%d",x);
   return 0 ;
   }
```

The value of x is

(a) 6
(b) 7
(c) 8
(d) 9

63%6/12

3/12

−2+11 = 9

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Toipc:Question

```
#include<stdio.h>
   int main(){
   int x;
   x=  2 * 3/4+4/4 +8-2+5/8;
   printf("%d",x);
   return 0 ;
   }
```

The value of x is

(a) 6
(b) 7
(c) 8
(d) 9

$= 6/4 + 1 + 8 - 2 + 0$

$= 1 + 1 + 8 - 2$

$= 8$ ✓

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Toipc:Question

```
#include<stdio.h>
   int main(){
   int x;
   x=  3/2*4+3/8+3;
   printf("%d",x);
   return 0 ;
   }
```

The value of x is

(a) 6
(b) 7
(c) 8
(d) 9

$3/2 * 4 + 3/8 + 3$

$1 * 4 + 0 + 3$

$= 4 + 0 + 3 = 7$

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Toipc:Question

```
#include<stdio.h>
   int main(){
   int x;
   x=  4+2%-8;
   printf("%d",x);
   return 0 ;
   }
```

The value of x is

(a) 6
(b) 7
(c) 8
(d) 9

Q=0

4+2%-8

4+2=6

4+2%-8 Q=0

4+2=6

Slide



Toipc: Comments in C

Comments are Not executable

Single line comment

// comment

/* This
is
multiline
Comment */

Raju.
written
a program
Raju Left the
company

Shyam
join the company

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Lexical Analysis - Remove the comments

for reference comments are used

//

/*

*/



Toipc: Scope of a variable

The scope of a variable in C is the block or the region in the program where a variable is declared, defined, and used. Outside this region, we cannot access the variable and it is treated as an undeclared identifier.



Toipc:Scope of variable

```
#include<stdio.h>
int  main {
     {
          int a =10 ;
          printf(“%d”; a); // will print the 10
     }


}
```

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Toipc:Scope of variable

```
#include<stdio.h>
int  main() {
   {
     int a =10 ;
   }
   printf("%d", a); // will not print 10
   return 0 ;
}
```



Toipc:Scope of variable

```
#include<stdio.h>
int  main() {
   {
     int a =10 ;
      printf("%d", a); // will print 10
   }

   return 0 ;
}
```



2 mins Summary

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THANK - YOU

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